

Ques. WHAT ARE THE DIFFERENT TYPES OF USB PORTS AND PLUGS AVAILABLE IN THE MARKET?

Talib Mirza

Ans. First appeared in 1996, universal serial bus (USB) is an industry standard that establishes specifications for cables and connectors, and protocols for connection, communication and power supply between computers, peripheral devices and other computers. The female connector mounted on the host or device is called receptacle (female), and the connector attached to the cable is called plug (male).

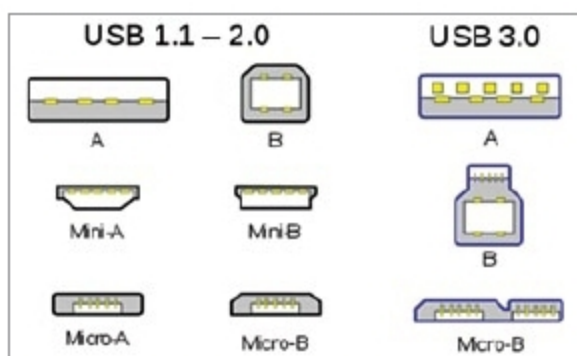


Fig. 1: USB connectors (Credit: Wikipedia.org)

There are three common types of USB connectors: standard, mini and micro. Standard is intended for desktop or portable equipment, mini is for mobile equipment and micro is for low-profile mobile equipment such as cellphones and tablets.

Again, based on functionality or version of the USB connector, there are four versions, namely, USB 1.0, USB 2.0, USB 3.0 and USB 4.0.

Based on physical design of the plugs and ports, there are many different types including USB Type-A, USB Type-B and USB Type-C.

USB Type-A. This is the standard, flat, rectangular interface that you find on one end of almost every USB cable. Most computers have multiple USB Type-A ports or receptors for connecting peripherals. You will also find these on game consoles, TVs and other devices.

USB Type-A mini. This type of connector was introduced in 2000 for use with smaller devices such as digital cameras, smartphones and tablet computers.

USB Type-A micro. The thinner micro-USB connector was introduced to replace the mini connector in devices manufactured in and after May 2007, including smartphones, personal digital assistants and cameras.

USB Type-B. This plug has a near-square cross-section with top exterior corners bevelled or having sloping edges. As part of a removable cable, it inserts into an upstream port on such device as a printer.

USB Type-B mini. Mini-B connector was the standard for transferring data to and from early smartphones and PDAs. Both Mini-A and Mini-B plugs are approximately 3mm x 7mm. You will also find these on digital cameras, external hard drives, USB hubs and other equipment.

USB Type-B micro. A micro-USB connector has a similar width as a mini-USB, but it is approximately half the thickness, enabling integration into thinner portable devices. Micro-USB was endorsed as the standard connector for data and power on mobile devices by cellular phone carrier groups, and is still a popular USB cable for smartphone devices.

USB Type-C. This fairly new standard was introduced in 2014 and is found in a limited number of devices such as Apple MacBooks (for charging, transferring data and playing videos). USB-C plug is reversible, and connects to both hosts and devices, replacing various Type-A and Type-B connectors and cables with a standard meant to be future-proof.

Q2. WHAT IS CONDUCTIVE PAINT? PLEASE PROVIDE LINKS WHERE IT IS AVAILABLE, ESPECIALLY IN INDIA?

Pamarthi Kanakaraja

A2. Conductive paint or ink is an ink that conducts electricity. It is typically created by infusing graphite or other conductive materials into ink. Conductive ink provides a cheaper way to create conductive traces or layouts as compared to traditional industrial standards such as etching copper from



Fig. 2: A typical conductive pen (Credit: bareconductive.com)

copper-plated substrates.

Conductive pens have been around for many years, and are used for repairing or adding traces to existing circuit boards on metal, glass, plastic or epoxy. For example, bare conductive paint is a multipurpose, electrically-conductive material perfect for DIY projects.

Silver conductive inks are used for printing RFID tags, circuit repair on PCBs, computer membrane keyboards, windshield defrosters, radio antennae in rear windows in cars and others.

Conductive paints are easily available in India. You can purchase these offline as well as online. Some of the e-commerce websites selling conductive paints are www.amazon.in, www.aliexpress.com, www.banggood.in and www.ebay.com

Q3. HOW CAN I REMOVE BLACK AND WHITE TEXT FROM A SILK-SCREEN OR COMPONENT SIDE OF THE PCB?

Preeti

A3. Black text from the PCB component side can be easily removed using a thinner or nail polish remover. A simple way is to soak a small piece of cloth/cotton in a thinner and rub on the text applying gentle force. The white legend or text cannot be easily removed using a commonly-available solvent, and you need another solution to remove it from the PCB.

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